

A COMPUTER-ASSISTED PROOF OF KUPERBERG'S SIX-CYLINDER CONJECTURE

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ABSTRACT. We prove that at most six pairwise non-overlapping infinite unit cylinders can simultaneously touch a unit ball. Six is achievable, so the answer to Kuperberg's question is exactly six. The proof is computer-assisted in the following sense: the statement is reduced to a finite case analysis with 2954 984 cases, and in each case a program verifies that one of three inequalities between explicit polynomials with rational coefficients holds throughout a box. Each case is an elementary arithmetic check, and the complete list of cases is provided with the paper.

1. INTRODUCTION

1.1. Formulation. Throughout, a *cylinder* means a closed solid right circular cylinder of infinite length and radius 1. Two cylinders are *non-overlapping* if their interiors are disjoint, and a cylinder *touches* the unit ball B if it meets B but not the interior of B . In 1990 W. Kuperberg asked how many pairwise non-overlapping cylinders can touch a fixed unit ball, and conjectured that the answer is six [5]. We prove that it is. Our proof has a computer-assisted component that does case analysis. The source code and the printout of cases can be found at [6].

Theorem 1.1. *At most six pairwise non-overlapping infinite unit cylinders can simultaneously touch a unit ball.*

The bound of Theorem 1.1 is attained and the answer to his question is exactly six, since the arrangements of six cylinders are easy to construct. One such configuration is constructed as follows. Let B be the unit ball centered at the origin, and let $A_1 A_2 \dots A_6$ be a regular hexagon of edge length 2 in the Oxy plane, centered at O . For $i = 1, \dots, 6$, let Z_i be the unit cylinder whose axis passes through A_i and is parallel to the Oz axis. Since each A_i lies at distance 2 from O and consecutive vertices are at distance 2 from each other, each Z_i touches B as well as its two neighboring cylinders, while the interiors of all seven bodies are mutually disjoint.

1.2. On the use of AI. Some of the ideas in this paper originated in work with Anthropic's Claude. The first proof we obtained that way was correct, but it drew at once on representation theory, harmonic analysis, spherical geometry and combinatorial optimization. Few readers hold that set of tools simultaneously. We reconstructed the argument to make a much simpler proof. What remains is an explicit table of integers, six finite checks on it, and one branch-and-bound computation. None of it need be taken on trust: neither the theory the data came from, nor our implementation of the computation.

The price is paid in Section 5, where matrices and weights appear with no motivation and no derivation. They are simply data, and every property the proof uses is established directly from them. The reasoning that produced them is sketched in Section 8, which is deliberately brief.

1.3. History of the problem and overview of the literature. Kuperberg posed the question at a DIMACS workshop in 1990, together with several arrangements of six unit cylinders touching a ball [5]. The first upper bound is due to Heppes and Szabó [4], who showed that at most eight cylinders can touch. Braß and Wenk [2] improved this to at most seven, by a short argument comparing areas on a concentric sphere. Whether seven is possible has been open since; Theorem 1.1 answers it.

Much of the work since has concerned the radius rather than the number. The familiar configuration of six unit cylinders is not rigid, and it was for a while natural to guess that six cylinders of radius greater than 1 could not touch the ball at all. Firsching [3] showed otherwise, finding by numerical exploration of the 18-dimensional configuration manifold six cylinders of radius 1.049659 that do. Ogievetsky and Shlosman [7] later exhibited a $\mathbb{D}_3$ -symmetric configuration with

$$r = \frac{1}{8}(3 + \sqrt{33}) \approx 1.093070,$$

and conjectured that this radius is the largest possible. In the opposite direction, Yardimci and Bezdek [9] proved that seven cylinders of radius 1.04965, which is Firsching's value, cannot touch the ball; at that one radius the analogue of Kuperberg's question is therefore settled. The case of radius exactly 1, which is the case Kuperberg asked about, is the one treated here.

The problem is a cousin of the kissing number question, where twelve unit balls can touch a unit ball and thirteen cannot [8]. Bounds of that kind are the natural home of Delsarte's linear programming method and of its semidefinite refinement [1], and it is a bound of that type, transported to the rotation group, that we use; Section 8 says more about where it comes from.

2. THE CONFIGURATION SPACE

Fix the unit ball B centered at O . Throughout, $u_0 = (0, 0, 1)$, $v_0 = (1, 0, 0)$, $a_1 = 2u_0 = (0, 0, 2)$.

A solid unit cylinder touches the ball B with center O if and only if the axis of the cylinder is tangent to the sphere $\mathbb{S}_2$ of radius 2. Two solid unit cylinders have disjoint interiors if and only if the distance between their axes is ≥ 2 .

Recall that $\text{SO}(3)$ is the group of all rotations of the three-dimensional Euclidean space around the origin. Every cylinder that touches B is uniquely determined by its axis $\ell = \{2u + tv : t \in \mathbb{R}\}$ with $u, v \in \mathbb{S}^2$, $u \perp v$; here $2u$ is the point of tangency and v a direction of the axis, unique up to sign. There exists a unique $g \in \text{SO}(3)$ such that $u = gu_0$, $v = gv_0$. We will write $\ell = g \cdot \ell_0$ with $\ell_0 = \{2u_0 + tv_0\}$.

Definition 2.1. We will define the *safe zone* $\mathcal{D}$ as a subset of $\text{SO}(3)$ in the following way

$$\mathcal{D} = \{g \in \text{SO}(3) : \text{dist}(\ell_0, g \cdot \ell_0) \geq 2\}.$$

Definition 2.2. A set $X \subseteq \text{SO}(3)$ will be called *non-overlapping* if $g^{-1}g' \in \mathcal{D}$ whenever g and g' are two distinct elements of X .

Consider N pairwise non-overlapping cylinders that touch B , with axes $\ell_1, \dots, \ell_N$, and choose $g_i \in \text{SO}(3)$ with $\ell_i = g_i \cdot \ell_0$. The g_i are distinct, since g determines $\ell = g \cdot \ell_0$. Rotations are isometries, so $\text{dist}(\ell_i, \ell_j) = \text{dist}(\ell_0, g_i^{-1}g_j \cdot \ell_0)$, and the cylinders having disjoint interiors is exactly the condition $g_i^{-1}g_j \in \mathcal{D}$ for all $i \neq j$. Thus $\{g_1, \dots, g_N\}$ is a non-overlapping subset of $\text{SO}(3)$ with N elements. Let $N_{\max}$ be the maximal number of elements that a non-overlapping subset of $\text{SO}(3)$ can have. Theorem 1.1 will be proved if we can establish $N_{\max} < 7$.

3. DELSARTE–BOCHNER SUPREMUM

Throughout this section, $L \in \mathbb{N}$ and, for $\ell = 1, \dots, L$, we fix

- (i) a symmetric positive-definite matrix $D_\ell \in \mathbb{R}^{n_\ell \times n_\ell}$, and
- (ii) a map $T^\ell : \text{SO}(3) \rightarrow \mathbb{R}^{n_\ell \times n_\ell}$ satisfying, for all $g_1, g_2, g \in \text{SO}(3)$,

$$T^\ell(g_1 g_2) = T^\ell(g_1) T^\ell(g_2) \quad \text{and} \quad T^\ell(g)^\top D_\ell T^\ell(g) = D_\ell.$$

Lemma 3.1. Under (i)–(ii), for every $g \in \text{SO}(3)$: $\det T^\ell(g) = \pm 1$; $T^\ell(e) = I$; and

$$T^\ell(g^{-1}) = T^\ell(g)^{-1} = D_\ell^{-1} T^\ell(g)^\top D_\ell.$$

Proof. Taking determinants in $T^\top D T = D$ gives $(\det T)^2 \det D = \det D$, and $\det D > 0$, so $\det T = \pm 1$; in particular every $T^\ell(g)$ is invertible. Then $T(e) = T(e \cdot e) = T(e)^2$ with $T(e)$ invertible forces $T(e) = I$, whence $T(g)T(g^{-1}) = T(e) = I$ gives $T(g^{-1}) = T(g)^{-1}$. Finally, multiplying $T^\top D T = D$ on the left by D^{-1} and on the right by T^{-1} gives $D^{-1} T^\top D = T^{-1}$. $\square$

Definition 3.2. For a sequence $\vec{C} = (C_\ell)_{\ell=1}^L$ of symmetric positive-semidefinite matrices $C_\ell \in \mathbb{R}^{n_\ell \times n_\ell}$, we define its Delsarte–Bochner supremum as

$$\text{db-sup}(\vec{C}) = \sup_{g \in \mathcal{D}} \sum_{\ell=1}^L \text{tr}(D_\ell^{-1} C_\ell T^\ell(g)).$$

Remark 3.3. When $D_\ell = \text{diag}(w_1^\ell, \dots, w_{n_\ell}^\ell)$ is diagonal and C_ℓ is symmetric, the correct pairing has the explicit entrywise form

$$\text{tr}(D_\ell^{-1} C_\ell T^\ell(g)) = \sum_{i,j} \frac{(C_\ell)_{ij}}{w_j^\ell} T^\ell(g)_{ij},$$

the sum of the entries of the Schur product of C_ℓ and $T^\ell(g)$, with the j -th column weighted by $1/w_j^\ell$. In particular $\text{tr}(D_\ell^{-1} C_\ell T^\ell(e)) = \text{tr}(D_\ell^{-1} C_\ell) = \sum_i (C_\ell)_{ii} / w_i^\ell$.

Theorem 3.4. Let $\vec{C} = (C_\ell)_{\ell=1}^L$ be a sequence of symmetric positive-semidefinite matrices $C_\ell \in \mathbb{R}^{n_\ell \times n_\ell}$. If $\text{db-sup}(\vec{C}) < 0$ and M is any real number with $\text{db-sup}(\vec{C}) \leq M < 0$, then

$$N_{\max} \leq 1 + \left\lfloor \frac{\sum_{\ell=1}^L \text{tr}(D_\ell^{-1} C_\ell)}{-M} \right\rfloor. \quad (1)$$

Proof. Let $g_1, \dots, g_N$ be distinct elements of a non-overlapping set (Definition 2.2), and put $S_\ell = \sum_{i=1}^N T^\ell(g_i)$. By Lemma 3.1,

$$T^\ell(g_i^{-1} g_j) = T^\ell(g_i)^{-1} T^\ell(g_j) = D_\ell^{-1} T^\ell(g_i)^\top D_\ell T^\ell(g_j),$$

so that, summing over all ordered pairs and using linearity and cyclicity of the trace,

$$\begin{aligned} \sum_{i=1}^N \sum_{j=1}^N \text{tr}(D_\ell^{-1} C_\ell T^\ell(g_i^{-1} g_j)) &= \sum_{i,j} \text{tr}(D_\ell^{-1} C_\ell D_\ell^{-1} T^\ell(g_i)^\top D_\ell T^\ell(g_j)) \\ &= \text{tr}\left((D_\ell^{-1} C_\ell D_\ell^{-1})(S_\ell^\top D_\ell S_\ell)\right) \geq 0, \end{aligned}$$

since the trace of a product of two symmetric positive-semidefinite matrices is non-negative (for such P, Q : $\text{tr}(PQ) = \text{tr}(\sqrt{Q} P \sqrt{Q}) \geq 0$, the trace of a positive-semidefinite matrix): $D_\ell^{-1} C_\ell D_\ell^{-1}$ is a congruence of C_ℓ , and $S_\ell^\top D_\ell S_\ell$ a congruence of D_ℓ , so both are symmetric positive semidefinite.

Summing over ℓ and separating the N diagonal pairs $i = j$ (where $T^\ell(e) = I$ gives the value $\sum_\ell \text{tr}(D_\ell^{-1} C_\ell)$) from the $N(N-1)$ off-diagonal pairs (where $g_i^{-1} g_j \in \mathcal{D}$ by Definition 2.2, so each term is at most $\text{db-sup}(\vec{C}) \leq M$),

$$0 \leq N \sum_{\ell=1}^L \text{tr}(D_\ell^{-1} C_\ell) + N(N-1)M.$$

Since $M < 0$, dividing by N and rearranging gives $(N-1)(-M) \leq \sum_\ell \text{tr}(D_\ell^{-1} C_\ell)$, which implies (1). $\square$

4. STRATEGY FOR THE PROOF

We will use the word *certificate* for an explicit choice of matrices D_ℓ , C_ℓ , and the maps T^ℓ for which we will be able to verify the conditions of Section 3 and of Theorem 3.4. Once those conditions are verified, the theorem gives the bound (1); for the certificate constructed below it gives $N_{\max} \leq 6$.

We will apply Theorem 3.4 with $L = 4$ and $M = -\frac{19}{20}$.

In Section 5 we construct, for $\ell = 1, \dots, 4$, the following:

- Explicit *rational* diagonal matrices $D_\ell = \text{diag}(w_1^\ell, \dots, w_{2\ell+1}^\ell)$ with positive diagonal entries,
- Explicit symmetric *rational* matrices C_ℓ , and
- Explicit maps $T^\ell : \text{SO}(3) \rightarrow \mathbb{R}^{(2\ell+1) \times (2\ell+1)}$.

The arguments of T^ℓ , which are the elements of $\text{SO}(3)$, will be represented with quaternions. We will prove:

- (F1) $T^\ell(g_1 g_2) = T^\ell(g_1) T^\ell(g_2)$ for all $g_1, g_2 \in \text{SO}(3)$;
- (F2) $T^\ell(g)^\top D_\ell T^\ell(g) = D_\ell$ for all $g \in \text{SO}(3)$;
- (F3) $C_\ell \succeq 0$.

Hypothesis (i) of Section 3 holds because each D_ℓ is diagonal with positive diagonal entries; facts (F1)–(F2) are hypothesis (ii); and (F3) is the hypothesis of Theorem 3.4. So the sequence $\vec{C} = (C_1, C_2, C_3, C_4)$ is admissible. Its trace sum is the rational number

$$\sum_{\ell=1}^4 \text{tr}(D_\ell^{-1} C_\ell) = \sum_{\ell=1}^4 \sum_i \frac{(C_\ell)_{ii}}{w_i^\ell} = \frac{23017}{4096}. \quad (2)$$

It then remains to bound the supremum $\text{db-sup}(\vec{C})$. By Remark 3.3, $\sum_\ell \text{tr}(D_\ell^{-1} C_\ell T^\ell(g))$ is an explicit polynomial with rational coefficients in the quaternion representing g , so the two final facts are:

- (F4) the safe zone $\mathcal{D}$ of Definition 2.1 is exactly the set of rotations satisfying two explicit polynomial inequalities $Q \geq 0$ and $U_1^2 \geq 4$ with integer coefficients (Section 6);
- (F5) on that semialgebraic set, the corresponding polynomial satisfies

$$\sum_{\ell=1}^4 \text{tr}(D_\ell^{-1} C_\ell T^\ell(g)) \leq -\frac{19}{20} \quad \text{for all } g \in \mathcal{D}. \quad (3)$$

The last claim will be established with a computer-assisted proof.

Granting (F1)–(F5), Theorem 3.4 with $M = -\frac{19}{20}$ and (2) gives

$$N_{\max} \leq 1 + \left\lfloor \frac{23017}{4096} \cdot \frac{20}{19} \right\rfloor = 1 + \left\lfloor \frac{460340}{77824} \right\rfloor = 1 + 5 = 6.$$

5. THE DATA, AND VERIFICATION THAT IT SATISFIES (F1)–(F3)

This section supplies the objects promised in Section 4. The data provided with the proof (see Appendix B) consists, for each $\ell = 1, \dots, 4$, of the following three components:

- positive rationals $w_1^\ell, \dots, w_{2\ell+1}^\ell$, forming the diagonal matrix $D_\ell = \text{diag}(w_1^\ell, \dots, w_{2\ell+1}^\ell)$;
- an integer table defining a $(2\ell+1) \times (2\ell+1)$ matrix $T_h^\ell(q)$ of polynomials in $q = (q_0, q_1, q_2, q_3)$, with coefficients in $\frac{1}{\text{Tden}_\ell} \mathbb{Z}$ for a deposited scale $\text{Tden}_\ell \in \{1, 2\}$ ($\text{Tden}_\ell = 2$ only for $\ell = 3$);
- a symmetric rational matrix $C_\ell \in \mathbb{Q}^{(2\ell+1) \times (2\ell+1)}$, deposited together with integer witnesses for its positive semidefiniteness.

Remark 5.1. No property of this data is assumed, and how it was found plays no role in the proof. It was obtained by Gram–Schmidt orthogonalisation of a basis of harmonic polynomials, as described in Section 8; neither the reader nor the verifier needs this. The checker sees only the provided tables, and must therefore establish the properties below directly.

Every property used is either proved here from the definitions or is one of the six exact computations (D1)–(D6), each carried out on the provided tables by the standalone checker of Appendix B.

5.1. Quaternions and rotations. For $q = (q_0, q_1, q_2, q_3) \in \mathbb{R}^4$ define

$$R(q) = \begin{pmatrix} q_0^2 + q_1^2 - q_2^2 - q_3^2 & 2(q_1q_2 - q_0q_3) & 2(q_1q_3 + q_0q_2) \\ 2(q_1q_2 + q_0q_3) & q_0^2 - q_1^2 + q_2^2 - q_3^2 & 2(q_2q_3 - q_0q_1) \\ 2(q_1q_3 - q_0q_2) & 2(q_2q_3 + q_0q_1) & q_0^2 - q_1^2 - q_2^2 + q_3^2 \end{pmatrix},$$

whose entries are integer quadratic forms in q , and write $|q|^2 = q_0^2 + q_1^2 + q_2^2 + q_3^2$. Let $p * q$ denote the Hamilton product,

$$\begin{aligned} (p * q)_0 &= p_0q_0 - p_1q_1 - p_2q_2 - p_3q_3, \\ (p * q)_1 &= p_0q_1 + q_0p_1 + p_2q_3 - p_3q_2, \\ (p * q)_2 &= p_0q_2 + q_0p_2 + p_3q_1 - p_1q_3, \\ (p * q)_3 &= p_0q_3 + q_0p_3 + p_1q_2 - p_2q_1. \end{aligned}$$

Lemma 5.2. *The following hold as polynomial identities in $\mathbb{Z}[p, q]$, verifiable by direct expansion:*

- $R(q)^\top R(q) = |q|^4 I$ and $\det R(q) = |q|^6$; in particular $R(q) \in \text{SO}(3)$ whenever $|q| = 1$;
- $R(p * q) = R(p) R(q)$ and $|p * q|^2 = |p|^2 |q|^2$;
- the ten products $q_a q_b$ ($0 \leq a \leq b \leq 3$) are $\mathbb{Z}$ -linear combinations of $|q|^2$ and the entries r_{ij} of $R(q)$; explicitly $4q_0^2 = |q|^2 + r_{11} + r_{22} + r_{33}$, $4q_1^2 = |q|^2 + r_{11} - r_{22} - r_{33}$, $4q_2^2 = |q|^2 - r_{11} + r_{22} - r_{33}$, $4q_3^2 = |q|^2 - r_{11} - r_{22} + r_{33}$, $4q_0q_1 = r_{32} - r_{23}$, $4q_0q_2 = r_{13} - r_{31}$, $4q_0q_3 = r_{21} - r_{12}$, $4q_1q_2 = r_{12} + r_{21}$, $4q_1q_3 = r_{13} + r_{31}$, $4q_2q_3 = r_{23} + r_{32}$.

Lemma 5.3. (a) *If q, q' are unit quaternions with $R(q) = R(q')$, then $q' = \pm q$.* (b) *Every $g \in \text{SO}(3)$ equals $R(q)$ for some unit quaternion q .*

Proof. (a) By Lemma 5.2(c) the entries of $R(q)$ together with $|q|^2 = 1$ determine every product $q_a q_b$, i.e. the symmetric rank-one matrix $qq^\top$. If $qq^\top = q'q'^\top$ then, taking any index a with $q_a \neq 0$, the a -th columns give $q_a q = q'_a q'$ with $q_a^2 = q'_a{}^2$, so $q' = \pm q$.

(b) First, every $g \in \text{SO}(3)$ has a fixed unit vector: since $\det g = 1$ and $g^\top g = I$,

$$\det(g - I) = \det(g) \det(I - g^\top) = \det((I - g)^\top) = \det(I - g) = (-1)^3 \det(g - I),$$

so $\det(g - I) = 0$ and there is a unit vector n with $gn = n$. On the plane $n^\perp$, g restricts to a rotation by some angle θ . Now set $q = (\cos \frac{\theta}{2}, n \sin \frac{\theta}{2})$, a unit quaternion; a direct computation from the displayed matrix shows $R(q)n = n$ and that $R(q)$ rotates $n^\perp$ by θ with the same orientation, so $R(q) = g$. $\square$

5.2. **The maps T^ℓ .** The maps T^ℓ are uniquely determined by the provided matrices $T_h^\ell(q)$, which are polynomials in (q_0, q_1, q_2, q_3) . The data satisfies the following relations, each of which is a finite exact computation:

(D1) every entry of $T_h^\ell(q)$ is homogeneous of degree 2ℓ ;

(D2) the identity

$$T_h^\ell(p * q) = T_h^\ell(p) T_h^\ell(q) \quad (4)$$

holds in $\mathbb{Q}[p_0, \dots, p_3, q_0, \dots, q_3]$;

(D3) the identity

$$T_h^\ell(q)^\top D_\ell T_h^\ell(q) = |q|^{4\ell} D_\ell \quad (5)$$

holds in $\mathbb{Q}[q_0, \dots, q_3]$;

(D4) $w_i^\ell > 0$ for every i ;

(D5) writing $\widehat{C}_\ell := 2^{20} C_\ell$ for the deposited integer matrix, and with the deposited positive integer scalar S_ℓ and the deposited integer matrix W_ℓ , the matrix

$$E_\ell = S_\ell \widehat{C}_\ell - W_\ell^\top W_\ell$$

is symmetric and diagonally dominant: $(E_\ell)_{ii} \geq \sum_{j \neq i} |(E_\ell)_{ij}|$ for every i ;

$$(D6) \sum_{\ell=1}^4 \sum_i \frac{(C_\ell)_{ii}}{w_i^\ell} = \frac{23017}{4096}.$$

Both (4) and (5) are identities between polynomials with integer coefficients after the deposited scales are cleared. Writing $A^\ell = \text{Tden}_\ell T_h^\ell$, which has integer entries, and $\widehat{D}_\ell = m_\ell D_\ell$, where m_ℓ is a common denominator of $w_1^\ell, \dots, w_{2\ell+1}^\ell$, they read

$$\text{Tden}_\ell A^\ell(p * q) = A^\ell(p) A^\ell(q), \quad A^\ell(q)^\top \widehat{D}_\ell A^\ell(q) = \text{Tden}_\ell^2 |q|^{4\ell} \widehat{D}_\ell,$$

so each amounts to comparing two polynomial expansions with integer coefficients, coefficient by coefficient. No approximation and no choice of inner product enters.

In the deposited data $S_\ell = 2^{26}$, and every entry of $S_\ell \widehat{C}_\ell$ and of $W_\ell^\top W_\ell$ is an integer of absolute value below 2^{46} , so (D5) is a check in exact 64-bit integer arithmetic. Note that diagonal dominance forces $(E_\ell)_{ii} \geq 0$, the right-hand side being a sum of absolute values.

Definition 5.4. For $g \in \text{SO}(3)$ set $T^\ell(g) = T_h^\ell(q)$, where q is any unit quaternion with $R(q) = g$; such a q exists by Lemma 5.3(b). The value does not depend on the choice: by Lemma 5.3(a) the only two choices are $\pm q$, and $T_h^\ell(-q) = T_h^\ell(q)$ by (D1), the degree 2ℓ being even.

Proposition 5.5. The matrices D_ℓ and the maps T^ℓ of Definition 5.4 satisfy hypotheses (i) and (ii) of Section 3.

Proof. By (D4) the matrix D_ℓ is diagonal with positive diagonal entries, hence symmetric positive definite; this is hypothesis (i).

For (F1), let $g_1, g_2 \in \text{SO}(3)$ and choose unit quaternions p, q with $R(p) = g_1$ and $R(q) = g_2$. By Lemma 5.2(b), $|p * q|^2 = |p|^2 |q|^2 = 1$ and $R(p * q) = R(p)R(q) = g_1 g_2$, so $p * q$ is a unit quaternion representing $g_1 g_2$. Evaluating (4) at (p, q) ,

$$T^\ell(g_1 g_2) = T_h^\ell(p * q) = T_h^\ell(p) T_h^\ell(q) = T^\ell(g_1) T^\ell(g_2).$$

For (F2), let $g \in \text{SO}(3)$ and choose a unit quaternion q with $R(q) = g$. Evaluating (5) at q and using $|q| = 1$,

$$T^\ell(g)^\top D_\ell T^\ell(g) = |q|^{4\ell} D_\ell = D_\ell.$$

Together these are hypothesis (ii). $\square$

5.3. **The matrices C_ℓ , and fact (F3).** The four matrices $C_1, \dots, C_4$ are also part of the certificate. Here they are simply given rational data; no motivation for their values is needed, and their construction is sketched in Section 8. Their positive semidefiniteness is not assumed: it follows from the deposited integer witnesses by the following two elementary facts.

Lemma 5.6. Let $E \in \mathbb{R}^{n \times n}$ be symmetric with $E_{ii} \geq \sum_{j \neq i} |E_{ij}|$ for every i . Then $E \succeq 0$.

Proof. Let $v \in \mathbb{R}^n$. Using $|v_i v_j| \leq \frac{1}{2}(v_i^2 + v_j^2)$ and then the symmetry of E ,

$$\left| \sum_{i \neq j} E_{ij} v_i v_j \right| \leq \sum_{i \neq j} |E_{ij}| \frac{1}{2}(v_i^2 + v_j^2) = \sum_i v_i^2 \sum_{j \neq i} |E_{ij}| \leq \sum_i E_{ii} v_i^2,$$

so $v^\top E v = \sum_i E_{ii} v_i^2 + \sum_{i \neq j} E_{ij} v_i v_j \geq 0$. $\square$

Proposition 5.7. *Each provided matrix C_ℓ is symmetric positive semidefinite, and*

$$\sum_{\ell=1}^4 \text{tr}(D_\ell^{-1} C_\ell) = \sum_{\ell=1}^4 \sum_{i=1}^{2\ell+1} \frac{(C_\ell)_{ii}}{w_i^\ell} = \frac{23017}{4096}. \quad (6)$$

Proof. By (D5) and Lemma 5.6, $E_\ell \succeq 0$. Also $W_\ell^\top W_\ell \succeq 0$, since $v^\top W_\ell^\top W_\ell v = |W_\ell v|^2 \geq 0$ for every v . Hence

$$2^{20} S_\ell C_\ell = S_\ell \widehat{C}_\ell = E_\ell + W_\ell^\top W_\ell \succeq 0,$$

and dividing by the positive scalar $2^{20} S_\ell$ gives $C_\ell \succeq 0$. Symmetry of C_ℓ is symmetry of the deposited $\widehat{C}_\ell$. The trace value is (D6). $\square$

5.4. The certificate polynomial.

Definition 5.8. The *certificate polynomial* is

$$P(q) = \sum_{\ell=1}^4 (|q|^2)^{4-\ell} \sum_{i,j} \frac{(C_\ell)_{ij}}{w_j^\ell} T_h^\ell(q)_{ij} \in \mathbb{Q}[q_0, q_1, q_2, q_3].$$

It is homogeneous of degree 8: by (D1) each $T_h^\ell(q)_{ij}$ is homogeneous of degree 2ℓ , and $2(4-\ell) + 2\ell = 8$ for every ℓ .

Proposition 5.9. *For every unit quaternion q , with $g = R(q) \in \text{SO}(3)$,*

$$P(q) = \sum_{\ell=1}^4 \text{tr}(D_\ell^{-1} C_\ell T^\ell(g)).$$

Consequently, fact (F5) of Section 4 is equivalent to

$$P(q) \leq -\frac{19}{20} \quad \text{for every unit quaternion } q \text{ with } R(q) \in \mathcal{D}. \quad (7)$$

Proof. For $|q| = 1$ the prefactors $(|q|^2)^{4-\ell}$ equal 1, and $T_h^\ell(q) = T^\ell(g)$ by Definition 5.4, so the claim is the entrywise formula of Remark 3.3 applied to the diagonal matrix D_ℓ . The equivalence with (F5) follows since every $g \in \mathcal{D}$ equals $R(q)$ for some unit quaternion q , by Lemma 5.3(b). $\square$

Remark 5.10. Although P has rational coefficients, clearing denominators makes it integral. Put $G_{\text{den}} = 40 \cdot 2^{20} = 41943040$; then all 35 coefficients of $G_{\text{den}} G$, where $G = -\frac{19}{20}|q|^8 - P$ is the cover polynomial of Section 7, are integers of absolute value $< 2^{32}$, hence exactly representable in double precision. A fingerprint identifies the polynomial: since $R(1, 0, 0, 0)$ is the identity, $P(1, 0, 0, 0)$ is the trace sum (2), so the coefficient of q_0^8 in $G_{\text{den}} G$ equals

$$-\left(\frac{19}{20} + \frac{23017}{4096}\right) G_{\text{den}} = -275539968.$$

Any implementation that reconstructs G from the deposited data must reproduce these 35 integers.

6. AN EXACT SEMIALGEBRAIC DESCRIPTION OF THE SAFE ZONE

The distance between two lines is not a polynomial function of their data, and it is not even Lipschitz near the locus where the two directions become parallel. We show that $\mathcal{D}$ is nonetheless a basic semialgebraic set, cut out by two explicit polynomial inequalities.

Fix $g \in \text{SO}(3)$ and write

$$d_1 = v_0, \quad d_2 = g v_0, \quad a_2 = 2g u_0, \quad w = a_2 - a_1, \quad n = d_1 \times d_2, \quad N_2 = |n|^2.$$

Thus ℓ_0 passes through a_1 with unit direction d_1 , and $g \cdot \ell_0$ passes through a_2 with unit direction d_2 . Writing $c = d_1 \cdot d_2$, we have $N_2 = 1 - c^2$, so $N_2 > 0$ exactly when d_1, d_2 are linearly independent, i.e. when the two lines are skew or intersecting, and $N_2 = 0$ exactly when they are parallel.

Lemma 6.1. Suppose $N_2 > 0$, and let $\bar{w}$ be the orthogonal projection of w onto $\text{span}\{d_1, d_2\}$. Then

$$N_2 |\bar{w}|^2 = (w \cdot d_1)^2 - 2(w \cdot d_1)(w \cdot d_2)(d_1 \cdot d_2) + (w \cdot d_2)^2. \quad (8)$$

Proof. Write $\bar{w} = s d_1 + t d_2$ and put $b_1 = w \cdot d_1$, $b_2 = w \cdot d_2$. Since $w - \bar{w}$ is orthogonal to d_1 and to d_2 , taking inner products with d_1, d_2 gives

$$\begin{pmatrix} 1 & c \\ c & 1 \end{pmatrix} \begin{pmatrix} s \\ t \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix},$$

whose determinant is $1 - c^2 = N_2 > 0$; Cramer's rule gives $N_2 s = b_1 - c b_2$ and $N_2 t = b_2 - c b_1$. Also $w - \bar{w} \perp \bar{w}$, so $|\bar{w}|^2 = w \cdot \bar{w} = s b_1 + t b_2$. Multiplying by N_2 and substituting,

$$N_2 |\bar{w}|^2 = (b_1 - c b_2) b_1 + (b_2 - c b_1) b_2 = b_1^2 - 2c b_1 b_2 + b_2^2. \quad \square$$

Definition 6.2. For $w \in \mathbb{R}^3$ and unit vectors d_1, d_2 set

$$\Gamma(w) = (w \cdot d_1)^2 - 2(w \cdot d_1)(w \cdot d_2)(d_1 \cdot d_2) + (w \cdot d_2)^2, \quad (9)$$

the right-hand side of (8). Thus $\Gamma(w) = N_2 |\bar{w}|^2$ whenever $N_2 > 0$, while Γ itself is a polynomial, defined for every configuration including the parallel one. Set further

$$Q = (|w|^2 - 4)N_2 - \Gamma(w), \quad U_1^2 = |w|^2 - (w \cdot d_1)^2. \quad (10)$$

Here U_1 is the distance from the point a_2 to the line ℓ_0 : that line passes through a_1 with unit direction d_1 , so the component of $w = a_2 - a_1$ along ℓ_0 has length $|w \cdot d_1|$ and the perpendicular component has squared length $|w|^2 - (w \cdot d_1)^2$. In the coordinates of Appendix A, N_2 and U_1^2 are polynomials of degree 2 in the entries of g and Q has degree 4; all have integer coefficients.

Lemma 6.3. With the notation above,

$$\text{dist}^2(\ell_0, g \cdot \ell_0) = \begin{cases} |w|^2 - \Gamma(w)/N_2, & N_2 > 0, \\ U_1^2, & N_2 = 0. \end{cases}$$

Moreover $\Gamma(w) = 0$ whenever $N_2 = 0$, and consequently

$$Q = N_2(\text{dist}^2(\ell_0, g \cdot \ell_0) - 4) \quad \text{when } N_2 > 0. \quad (11)$$

Proof. Case $N_2 > 0$. The distance between the two lines is the distance from w to $\text{span}\{d_1, d_2\}$, namely $|w - \bar{w}|$, since $\{a_2 + t d_2\} - \{a_1 + s d_1\} = w + t d_2 - s d_1$ ranges over $w + \text{span}\{d_1, d_2\}$. Hence $\text{dist}^2 = |w|^2 - |\bar{w}|^2 = |w|^2 - \Gamma(w)/N_2$ by Lemma 6.1. Multiplying by N_2 and subtracting $4N_2$ gives $N_2 \text{dist}^2 - 4N_2 = (|w|^2 - 4)N_2 - \Gamma(w) = Q$, which is (11).

Case $N_2 = 0$. Then $d_2 = \pm d_1$, so $w \cdot d_2 = \pm(w \cdot d_1)$ and $d_1 \cdot d_2 = \pm 1$ with matching signs; substituting into (9),

$$\Gamma(w) = (w \cdot d_1)^2 - 2(w \cdot d_1)^2 + (w \cdot d_1)^2 = 0.$$

Also $t d_2 - s d_1 = (\pm t - s) d_1$ ranges over $\text{span}\{d_1\}$, so the distance between the lines is the distance from w to $\text{span}\{d_1\}$, which is U_1 . $\square$

Theorem 6.4. The safe zone $\mathcal{D}$ satisfies $\mathcal{D} = \{Q \geq 0\} \cap \{U_1^2 \geq 4\}$.

Proof. Fix g and write $\delta = \text{dist}(\ell_0, g \cdot \ell_0)$, so that $g \in \mathcal{D}$ means $\delta \geq 2$. Since a_2 lies on $g \cdot \ell_0$, we have $\delta \leq U_1$ always.

Case $N_2 > 0$. By (11) and $N_2 > 0$, the sign of Q is the sign of $\delta^2 - 4$; hence $Q \geq 0 \iff \delta \geq 2$. If $Q \geq 0$ then $U_1 \geq \delta \geq 2$, so the second inequality is automatic. Therefore, on $\{N_2 > 0\}$, $\mathcal{D} = \{Q \geq 0\} = \{Q \geq 0\} \cap \{U_1^2 \geq 4\}$.

Case $N_2 = 0$. Then $\Gamma(w) = 0$ by Lemma 6.3 and $N_2 = 0$, so $Q = 0$: the first inequality holds identically and carries no information. By Lemma 6.3 again $\delta = U_1$, so $\delta \geq 2 \iff U_1^2 \geq 4$. Therefore, on $\{N_2 = 0\}$, $\mathcal{D} = \{U_1^2 \geq 4\} = \{Q \geq 0\} \cap \{U_1^2 \geq 4\}$. $\square$

Remark 6.5. Neither inequality can be dropped. On the parallel locus $\{N_2 = 0\}$ the first is vacuous, and $\{Q \geq 0\}$ alone would admit antiparallel cylinders at any separation, including deeply overlapping ones; the second inequality removes exactly those. Off that locus the second is implied by the first. This is the algebraic resolution of the parallel-axis degeneracy, which is what obstructs a naive treatment: the parallel locus has measure zero, so a numerical search that samples the skew region will not see it.

We use the homogenisations: on the unit sphere $\{|q|^2 = 1\}$ the conditions $Q \geq 0$ and $U_1^2 \geq 4$ are equivalent to $Q_h(q) \geq 0$ and $U_h(q) \geq 0$, where Q_h (degree 8) and U_h (degree 4) are the homogeneous integer polynomials obtained from (10) by substituting the integer quadratic forms $R(q)_{ij}$ of Section 5.1 for the entries of g and replacing each constant by the matching power of $|q|^2$. They are written out in Appendix A.

7. A COMPUTER-ASSISTED PROOF OF FACT (F5).

We certify (7). Define the homogeneous degree-8 polynomial

$$G(q) = -\frac{19}{20}|q|^8 - P(q), \quad (12)$$

which has integer coefficients after the scaling of Remark 5.10. By Theorem 6.4 and the homogenisation above, (7) is equivalent to

$$G(q) \geq 0 \quad \text{on} \quad K = \{q \neq 0 : Q_h(q) \geq 0 \text{ and } U_h(q) \geq 0\}, \quad (13)$$

since G, Q_h, U_h are homogeneous, so membership in K and the sign of G are constant along rays, and on unit quaternions (13) reads $P(q) \leq -\frac{19}{20}$ for $R(q) \in \mathcal{D}$. Because these polynomials are homogeneous and even in q , it suffices to certify (13) on the four cube-faces $\{q_i = 1, |q_j| \leq 1 \ (j \neq i)\}$, $i = 0, \dots, 3$, whose rays cover all of $\mathbb{R}^4 \setminus \{0\}$ up to $q \mapsto -q$.

7.1. Algorithm. For each $i \in \{0, 1, 2, 3\}$, we make a separate problem. Set $q_i = 1$, and re-write G, Q_h , and U_h as polynomials in three variables. In each problem, we can re-label these variables and call them x, y , and z . The theorem will be proved if we manage to partition $[-1, 1]^3$ into finitely many boxes of the form $[L_x, R_x] \times [L_y, R_y] \times [L_z, R_z]$, where none of the open intervals $(L_x, R_x), (L_y, R_y), (L_z, R_z)$ contains 0, and to prove that on each box at least one of the following three inequalities is satisfied:

$$U_h(x, y, z) < 0, \quad Q_h(x, y, z) < 0, \quad \text{or} \quad G(x, y, z) \geq 0. \quad (14)$$

This is a technique for proving that a polynomial $P(x, y, z)$ has a constant sign on $[L_x, R_x] \times [L_y, R_y] \times [L_z, R_z]$. First, split P into sum of monomials

$$P(x, y, z) = \sum_j \gamma_j x^{a_j} y^{b_j} z^{c_j}.$$

Then, for each j , the maximum M_j and the minimum m_j of the monomial $\gamma_j x^{a_j} y^{b_j} z^{c_j}$ is attained at a corner of the box: each factor is monotone on its interval, because 0 is not interior to it. Hence, m_j and M_j are possible to calculate manually, or by a computer. Define $M_P = \sum_j M_j$ and $m_P = \sum_j m_j$. If it happens that $M_P < 0$, then we can guarantee that $P(x, y, z) < 0$ on the box $[L_x, R_x] \times [L_y, R_y] \times [L_z, R_z]$. Similarly, if it happens that $m_P \geq 0$, then we are allowed to conclude that $P(x, y, z) \geq 0$.

None of this requires rounding. After the scaling of Remark 5.10 the coefficients γ_j are integers, and every corner produced by repeated halving has the form $k/2^d$ with k, d integers, so each M_j and m_j , and hence M_P and m_P , is a rational number whose denominator is a power of two. Both tests ask only for a sign, and a sign is unchanged by multiplication by a positive number: multiplying through by 2^{8d} , the total degree being 8, turns each of them into a comparison between two integers. At the depths reached here those integers stay below 2^{120} .

The deposited program does not take that route: it evaluates M_P and m_P in double precision. This is nevertheless enough, because the margins are wide. Write $u = 2^{-53}$ for the unit roundoff and $\gamma_N = Nu/(1 - Nu)$. Forming one m_j costs at most eleven multiplications, and summing the 35 of them costs 34 additions, so the computed value of m_P differs from the exact one by at most $\gamma_{50} \sum_j |m_j| < 5.6 \times 10^{-15} \sum_j |m_j|$, and the same bound holds for M_P . Over the whole run the smallest value of $m_P / \sum_j |m_j|$ at a certified box is 1.3×10^{-7} , and the smallest value of $-M_P / \sum_j |M_j|$ at a discarded box is 1.3×10^{-8} . Each exceeds the error bound by a factor greater than 10^6 , so every sign decided during the run is decided correctly.

There is one technical refinement of the algorithm that doesn't require a very large number of equally-sized boxes. We first split the $[-1, 1]^3$ into 8 boxes by midpoints, after which 0 is always an endpoint and never interior. Then for each box we try to see if one of (14) is satisfied. If it is, we say that the box is certified. If it is not, then we split the box into 8 equally sized smaller boxes. We keep going until the number of uncertified boxes becomes 0. Nothing guarantees in advance that this happens: if some box could never be certified, the splitting would continue forever and we would have proved nothing. That it does stop is the whole content of the computation. In our case, the process was finite.

The above algorithm with target $-\frac{19}{20}$ as in (12) terminates. About 3.4 million boxes are tested in all: 1 766 376 of them are certified, 1 188 608 are discarded, and the rest are split further. No box remains, and none fails. The boxes that are never split are the 2 954 984 cases referred to in the abstract; they are what the four partitions consist of, and each is settled by one of the three inequalities (14). Of the 1 188 608 discards, 96 364 are by $U_h < 0$ and 1 092 244 by $Q_h < 0$, so almost all of the discarding is done by Q_h ; this is the quantitative form of the remark in Section 6 that the second inequality does its work only near the parallel locus.

The subdivision does not go deep: the smallest boxes produced have half-width 2^{-10} .

This completes the proof. Facts (F1)–(F5) are all established, so $N_{\max} \leq 6$ by Section 4. By Section 2 any N pairwise non-overlapping unit cylinders touching the unit ball give a non-overlapping subset of $\text{SO}(3)$ with N elements, so $N \leq N_{\max} \leq 6$. This is Theorem 1.1.

8. WHERE THE CERTIFICATE COMES FROM

Nothing in this section is used anywhere in the proof. It is included because the data of Section 5 would otherwise arrive with no explanation at all, and because a reader who wants to construct a certificate of their own will need it. A reader who only wants to check the theorem may stop at Section 7.

8.1. The bound. By Section 2, a configuration of non-overlapping touching cylinders is an independent set in the graph on $\text{SO}(3)$ in which g and g' are adjacent when $g^{-1}g' \notin \mathcal{D}$. Bounding independence numbers of graphs invariant under a group action is the subject of Delsarte's linear programming method and of its semidefinite refinement; on a compact group the objects that produce such bounds are the functions of positive type. Theorem 3.4 is one such bound, stated in the narrow form we need so that it can be proved in half a page. The general framework, and its use for packing problems on spheres and on $\text{SO}(3)$, is that of Bachoc and Vallentin [1].

8.2. Why sums of traces of representations. The irreducible real representations of $\text{SO}(3)$ are its actions on the spaces H_ℓ of harmonic polynomials on $\mathbb{R}^3$, homogeneous of degree ℓ , of dimension $2\ell + 1$. Bochner's theorem for compact groups identifies the continuous functions of positive type on $\text{SO}(3)$ as the sums

$$g \mapsto \sum_{\ell \geq 0} \text{tr}(A_\ell \pi^\ell(g)), \quad A_\ell \succeq 0,$$

over those representations π^ℓ . Truncating the sum at $\ell \leq L$ turns the search for a good bound into a finite semidefinite program in the matrices A_ℓ . The maps T_h^ℓ of Section 5.2 are these representations written in an explicit basis, and the matrices C_ℓ are the corresponding A_ℓ . Theorem 3.4 itself uses none of this: it assumes only the two identities (i)–(ii) of Section 3, which is why those, and not irreducibility or continuity, are what Section 5 verifies.

8.3. The basis, and why the data is rational. Fix ℓ and take any basis of H_ℓ with integer coefficients. Apply Gram–Schmidt with respect to

$$\langle p, r \rangle = \frac{1}{4\pi} \int_{\mathbb{S}^2} p r \, d\sigma.$$

Every such inner product is rational, due to

$$\frac{1}{4\pi} \int_{\mathbb{S}^2} x^a y^b z^c \, d\sigma = \begin{cases} \frac{(a-1)!!(b-1)!!(c-1)!!}{(a+b+c+1)!!}, & a, b, c \text{ all even,} \\ 0, & \text{otherwise,} \end{cases}$$

which is easy to prove (this is a standard integral and can be calculated using spherical coordinates). Hence, the process never leaves $\mathbb{Q}$. Clearing denominators gives integer polynomials $h_1^\ell, \dots, h_{2\ell+1}^\ell$. The Gram matrix is then diagonal by construction, and it is the matrix D_ℓ of Section 5: $\langle h_i^\ell, h_j^\ell \rangle = \delta_{ij} w_i^\ell$ with $w_i^\ell > 0$.

Since $\text{SO}(3)$ acts on H_ℓ , for each $g = R(q)$ there are unique coefficients with

$$h_j^\ell(R(q)^\top x) = \sum_{i=1}^{2\ell+1} T_h^\ell(q)_{ij} h_i^\ell(x), \tag{15}$$

and these are the deposited tables. From (15) the three properties (D1)–(D3) are immediate: homogeneity of degree 2ℓ is a degree count, since the entries of $R(q)$ are quadratic in q ; (D2) is the composition law of the action; and (D3) is the invariance of σ under $\text{SO}(3)$, which makes each g act by an isometry of $\langle \cdot, \cdot \rangle$. We stress that the proof does not proceed this way. The harmonic polynomials h_i^ℓ are not deposited and the verifier never sees them; Section 5 establishes (D1)–(D3) directly from the tables, so no property of the construction above has to be trusted.

8.4. The matrices C_ℓ . The C_ℓ play the role of the matrices A_ℓ above: they are the free variables of the semidefinite program, and the bound that Theorem 3.4 returns depends on them through the ratio of $\text{db-sup}(\vec{C})$ to the trace sum $\sum_\ell \text{tr}(D_\ell^{-1} C_\ell)$. The deposited values are dyadic, with denominators dividing 2^{20} .

A numerical solution of a semidefinite program is not usable as it stands, and rounding a feasible point need not leave it feasible. The paper therefore takes nothing at all from the computation that produced these matrices: positive semidefiniteness is re-established on the rounded values by the integer witnesses of (D5), the trace sum by (D6), and the bound on $\mathcal{D}$ by the computation of Section 7.

8.5. The two constants. The truncation $L = 4$ and the threshold $M = -\frac{19}{20}$ are the only arbitrary-looking choices in Section 4, and neither is delicate.

Larger L gives a richer family of positive-type functions and hence a bound at least as good; $L = 4$ is what we found sufficient.

For M there is a whole interval of admissible values. Any M with $\text{db-sup}(\vec{C}) \leq M < 0$ may be used, and by (1) the resulting bound is 6 as soon as $\frac{23017}{4096}/(-M) < 6$, that is, as soon as $-M > \frac{23017}{24576} = 0.93656\dots$. On the other side, M must be at least $\text{db-sup}(\vec{C})$. The rational point $q^* = (1, \frac{2365}{4096}, 0, 0)$ lies in the cone K of Section 7 — exactly, $U_h(q^*) = 55008914419/2^{46} > 0$ and $Q_h(q^*) = 0$ — and $P(q^*)/|q^*|^8 = -0.98406\dots$ by exact rational evaluation, so $\text{db-sup}(\vec{C}) \geq -0.98406\dots$ and no M below that value can be used; numerically $\text{db-sup}(\vec{C}) \approx -0.9838$. So any

$$M \in [\text{db-sup}(\vec{C}), -0.93656\dots) \quad \text{with } \text{db-sup}(\vec{C}) \approx -0.9838,$$

yields the theorem, and $-\frac{19}{20} = -0.95$ is a convenient rational lying inside it with room on both sides. The cover of Section 7 certifies the left-hand requirement; the arithmetic of Section 4 certifies the right-hand one.

APPENDIX A. THE DOMAIN POLYNOMIALS IN COORDINATES

With $g = (g_{ij})$, $u = (g_{13}, g_{23}, g_{33})$, $d_2 = (g_{11}, g_{21}, g_{31})$, $w = (2g_{13}, 2g_{23}, 2g_{33} - 2)$, $d_1 = (1, 0, 0)$, $n = (0, -g_{31}, g_{21})$:

$$N_2 = g_{21}^2 + g_{31}^2, \quad w \cdot d_1 = 2g_{13}, \quad w \cdot d_2 = 2g_{13}g_{11} + 2g_{23}g_{21} + (2g_{33} - 2)g_{31}, \quad d_1 \cdot d_2 = g_{11},$$

$$U_1^2 = 4g_{23}^2 + (2g_{33} - 2)^2, \quad Q = (|w|^2 - 4)N_2 - \Gamma(w),$$

with Γ as in (9). In a unit quaternion each g_{ij} is the integer quadratic form $G_{ij} = R(q)_{ij}$ of Section 5.1, making U_1^2 , N_2 degree 4 and Q degree 8. The homogenisations used in Section 7 are the following explicit integer forms. Set

$$\mathbb{W} = (2G_{13}, 2G_{23}, 2G_{33} - 2|q|^2), \quad \mathbb{D}_2 = (G_{11}, G_{21}, G_{31}),$$

the homogenisations of w and d_2 (entries homogeneous of degree 2; in w only the third entry changes, its constant -2 becoming $-2|q|^2$), and define

$$U_h = |\mathbb{W}|^2 - (\mathbb{W} \cdot d_1)^2 - 4|q|^4 = |\mathbb{W}|^2 - 4G_{13}^2 - 4|q|^4,$$

$$Q_h = (|\mathbb{W}|^2 - 4|q|^4)(G_{21}^2 + G_{31}^2) - \left[|q|^4 (\mathbb{W} \cdot d_1)^2 - 2G_{11} (\mathbb{W} \cdot d_1)(\mathbb{W} \cdot \mathbb{D}_2) + (\mathbb{W} \cdot \mathbb{D}_2)^2 \right],$$

where $\mathbb{W} \cdot d_1 = 2G_{13}$ and $\mathbb{W} \cdot \mathbb{D}_2 = 2G_{13}G_{11} + 2G_{23}G_{21} + (2G_{33} - 2|q|^2)G_{31}$. Every constant of the inhomogeneous expressions has been replaced by the matching power of $|q|^2$; note the weights are *not* uniform: inside the bracket (the homogenisation of $\Gamma(w)$ from (9)) only the term $(\mathbb{W} \cdot d_1)^2$, of degree 4, carries the factor $|q|^4$, the other two terms being of degree 8 already, while $N_2 = G_{21}^2 + G_{31}^2$ is homogeneous as it stands and is left unchanged.

Lemma A.1.

- (a) U_h and Q_h are homogeneous of degrees 4 and 8 with integer coefficients, and are invariant under $q \mapsto -q$.
- (b) For every unit quaternion q , with $g = R(q) \in \text{SO}(3)$,

$$U_h(q) = U_1^2(g) - 4 \quad \text{and} \quad Q_h(q) = Q(g).$$

- (c) For every $q \neq 0$, writing $\hat{q} = q/|q|$ and $\hat{g} = R(\hat{q})$,

$$U_h(q) \geq 0 \iff U_1^2(\hat{g}) \geq 4 \quad \text{and} \quad Q_h(q) \geq 0 \iff Q(\hat{g}) \geq 0.$$

Hence, under the parametrisation of Lemma 5.3, the safe zone of Theorem 6.4 corresponds exactly to the cone $K = \{q \neq 0 : Q_h(q) \geq 0, U_h(q) \geq 0\}$ of Section 7.

- (d) U_h and Q_h are the unique homogeneous polynomials of their degrees satisfying (b).

Proof. (a) Each G_{ij} and $|q|^2$ is an integer quadratic form, so every entry of $\mathbb{W}$ and $\mathbb{D}_2$ is homogeneous of degree 2; counting degrees termwise, each displayed summand of U_h has degree 4 and each summand of Q_h degree 8. A homogeneous polynomial of even degree is invariant under the antipode.

(b) Fix $|q| = 1$, so $|q|^2 = 1$, $|q|^4 = 1$, and $g = R(q) \in \text{SO}(3)$ by Lemma 5.2(a). Then $\mathbb{W} = w$ and $\mathbb{D}_2 = d_2$: the homogenisation is invisible on the unit sphere. Substituting, each homogenised expression restricts to its Section 6 counterpart: $U_h = |w|^2 - (w \cdot d_1)^2 - 4 = U_1^2 - 4$, and $Q_h = (|w|^2 - 4)N_2 - \Gamma(w) = Q$ by (10).

(c) By (a), $U_h(q) = |q|^4 U_h(\hat{q})$ and $Q_h(q) = |q|^8 Q_h(\hat{q})$ with $|q|^4, |q|^8 > 0$, so the signs are constant along rays; now apply (b) at $\hat{q}$. The final statement combines this with Theorem 6.4 and Lemma 5.3(b).

(d) If h_1, h_2 are homogeneous of the same degree d and agree on the unit sphere, then $h_1(q) = |q|^d h_1(\hat{q}) = |q|^d h_2(\hat{q}) = h_2(q)$ for all $q \neq 0$, hence everywhere. So (b) determines U_h and Q_h uniquely: there is no normalisation freedom, so any implementation that constructs homogeneous forms of degrees 4 and 8 restricting to $U_1^2 - 4$ and Q on the unit sphere must produce exactly the expansions displayed above, coefficient for coefficient. $\square$

APPENDIX B. DEPOSITED DATA AND STANDALONE VERIFIER

The certificate is deposited as a single plain-text file in a self-describing format, designed to be read by a short standalone parser. Each record consists of a key, written `[key]Name[/key]`, followed by a value block `[value] ... [/value]`. A value block holds one or more rows; each row, and each entry within a row, is delimited by `{` and `}`. Every value is therefore a nested list of integers of depth two, including those that carry a single number, which appear as one row with one entry. Every key ends in a four-digit index `000ℓ`.

The certificate consists of 28 records, seven for each $\ell = 1, \dots, 4$:

- `WNum000ℓ` and `WDen000ℓ`: the weights w_i^ℓ as numerator/denominator pairs, forming

$$D_\ell = \text{diag}(w_1^\ell, \dots, w_{2\ell+1}^\ell);$$

- `Tpoly000ℓ` together with the scale `Tden000ℓ`: rows $i \ j \ e_0 \ e_1 \ e_2 \ e_3 \ c$, the entry $T_h^\ell(q)_{ij}$ being the sum of $(c/\text{Tden}_\ell) q_0^{e_0} q_1^{e_1} q_2^{e_2} q_3^{e_3}$ over the rows carrying that i, j ;
- `Cscaled000ℓ` = $2^{20} C_\ell$, as the rows of an integer matrix;
- `Swit000ℓ` and `Wwit000ℓ`: the integer scalar S_ℓ and the integer matrix W_ℓ of (D5).

Nothing further is deposited. Every other quantity in the proof is derived from these 28 records: the reciprocals $1/w_i^\ell$ from `WNum/WDen`; the certificate polynomial P of Definition 5.8, and hence the cover polynomial G of (12), from the C_ℓ , the T_h^ℓ tables and the weights; and U_h, Q_h from the matrix $R(q)$ of Section 5.1 by the formulas of Appendix A. The two rational constants of the argument, the threshold $-\frac{19}{20}$ and the trace sum $\frac{23017}{4096}$, are those of Section 4.

On these records the standalone verifier confirms:

- (D1) and (D4), by inspection of the exponents in `Tpoly000ℓ` and of the signs in `WNum000ℓ, WDen000ℓ`;
- (D2) and (D3), by expanding both sides of (4) and (5) and comparing them term by term;
- (D5), forming $E_\ell = S_\ell(2^{20} C_\ell) - W_\ell^\top W_\ell$ and testing symmetry and diagonal dominance; every intermediate product is an integer of absolute value below 2^{46} , so this runs in 64-bit integer arithmetic;
- (D6), by evaluating $\sum_\ell \sum_i (C_\ell)_{ii} / w_i^\ell$ and testing that 4096 times the result equals 23017;
- the subdivision of Section 7, reproducing the counts reported there: every box certified or discarded, and no failures.

The witness check (D5) runs entirely in 64-bit integer arithmetic. The remaining checks are carried out in double precision, yet they are exact rather than approximate. Every number entering them is a dyadic rational $m/2^k$ with $|m| < 2^{53}$: the table entries c/Tden_ℓ with $\text{Tden}_\ell \in \{1, 2\}$, the reciprocals $1/w_i^\ell$, which are dyadic in the deposited data, and the entries $2^{-20}(2^{20} C_\ell)$ of C_ℓ ; and every intermediate sum and product formed in (D2), (D3) and (D6) is again of this form. Such numbers are represented exactly in IEEE double precision, and sums and products of them incur no rounding, so each comparison above is a comparison of exact rational values. Rounding enters only in the subdivision of Section 7, where the sign of every decision is protected by the margin argument given there.

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